

the data and thus, a fast CPU with slow RAM would not make an impressive pair.

2. **Instruction Sets:** depending on what you want to run on the CPU, the instruction set matters. There are main instruction sets like the x86 (32-bit instruction set) or amd64/x86_64 (the 64-bit instruction set) and then there are others like SSE4.1 (the SIMD instruction set implemented on Intel chips). There may be dedicated instruction sets for encryption and decryption too (e.g. the AES Instruction Set on Intel and AMD CPUs), so that encrypted data can be read and processed as fast as possible.
3. **Speed and Energy balance:** The faster the processor works, the more energy it consumes. However, we do not use that much power most of the time. For example, web-browsing or document editing is not very compute-intensive. Many modern CPUs come with speed enhancement technologies. One with such a technology would be better than another which cannot increase or decrease the frequency on demand.
4. **Cache Size and Levels:** Cache is the fastest memory there is in a computer system. This memory is packaged into the CPU and resides very close to the compute cores. It is also very expensive and cannot be upgraded or changed because it is inside the processor chip. No matter what the CPU is executing, it brings the data (and the instructions) into the cache before working on it. This is so because cache is the only memory accessible to the CPU. Every time the CPU wants to fetch some data, whether it be an instruction or data, it looks for the same in the cache. If the data is there (called a cache hit), it loads the data into an appropriate register (or instruction pipeline, if it is an instruction) and then executes it. If the data is not present in the cache (cache miss), it looks for it in the RAM and loads it from there. It is noteworthy that caches are very fast memory units designed to be at par with the processing speed of the CPU itself. RAM modules on the other hand are much slower than CPUs. A cache miss usually triggers data transfer from RAM. While the data is being loaded into the cache, the CPU cannot perform that task and thus may stay idle for that duration. It goes without saying that bigger the cache, faster your computer is going to feel!
5. **Threading and Cores:** If there's anything that gets compared as much as a processor's clock frequency, it's the number of cores. It might sound funny but your OS might be deceiving you a little. The Task Manager of your OS might show 8 cores, while there are only 4 (as you can see